# TPG4190 Seismic data acquisition and processing
# Lecture 6: Processing

B. Arntsen

NTNU
Department of Geoscience and petroleum
borge.arntsen@ntnu.no

Trondheim fall 2025

# Overview

- Basic processing sequence
- Velocity analysis
- Migration

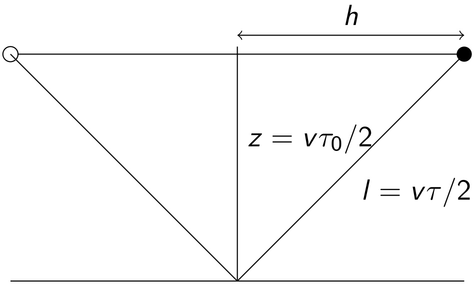
# Basic processing sequence

1. Input data
2. Velocity analysis
3. NMO + Stack
4. Output result

# Basic+ processing sequence

1. Input data
2. Velocity analysis
3. Migration
4. Output result

# NMO and Stack




The traveltime $\tau(h)$ is:

$$l^2 = z^2 + h^2 \tag{1}$$

which gives by inserting $v\tau/2$ for $l$ and $v\tau_0/2$ for $z$

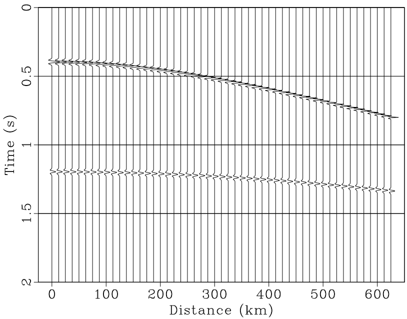
$$\tau(h) = \sqrt{\tau_0^2 + 4h^2/v^2}. \tag{2}$$

# NMO and Stack

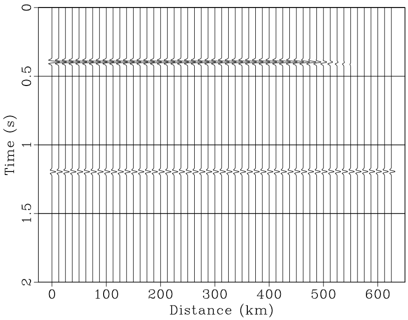
Nmo-correction:

$$\Delta \tau = \tau(h) - \tau_0, \tag{3}$$

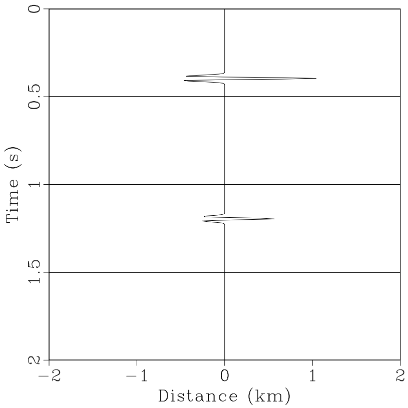
# Cmp



# Nmo



# Stack



# NMO and Stack

Average velocity $v_{rms}$ defined by

$$v_{rms}^2(t_0) = \frac{1}{t_0}\int_0^{t_0} v^2(t)dt \tag{4}$$

$v(t)$ : Interval velocity. The traveltime equation (4) then becomes

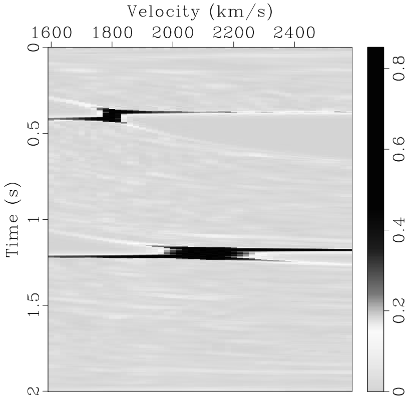
$$\tau(h) = \sqrt{t_0^2 + 4h^2/v_{rms}^2(t_0)}. \tag{5}$$

# Velocity analysis

Estimate $v_{rms}$ from equation (4)

1. Make a guess of $v$
2. Perform Nmo correction for all $t_0$ using guess of $v$
3. Make a stack trace
4. Perform above steps for a range of guesses of $v$
5. Plot all stack traces in a velocity spectrum

# Velocity spectrum



# Semblance

Stack is usually replaced with semblance to get better velocity spectra

$$S(t) = \frac{\int_0^{h_{max}} dh\, p^2(t,h)}{\int_0^T dt \int_0^{h_{max}} dh\, p^2(t,h)} \qquad (6)$$

- $p(t,h)$ : data
- $h_{max}$: maximum offset.

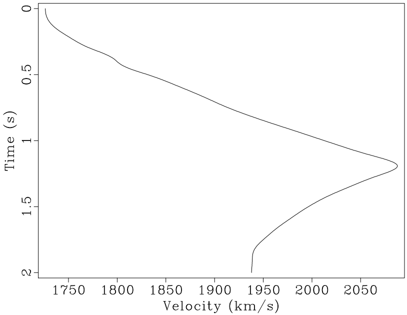
This equation can not be used directly. Why?

# Semblance

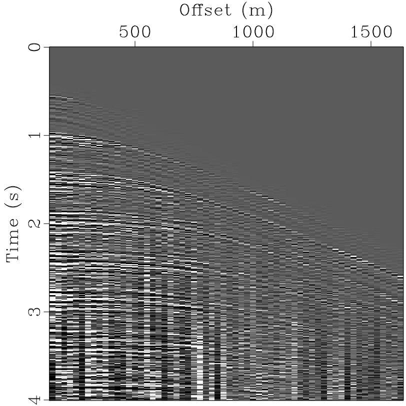
$$S_k = \frac{\sum_{i=0}^{N_h} p_{k,i}^2}{\sum_{k=0}^{N_t} \sum_{i=0}^{N_h} p_{k,i}^2} \tag{7}$$

- $S_k = S(t = k\Delta t), k = 0, \ldots, N_t$
- $p_{k,i} = p(t = k\Delta t, h = i\Delta h), i = 0, \ldots, N_h$.
- $\Delta t$ time sampling interval,
- $\Delta h$ distance between offsets
- $N_t$,$N_h$ : No of time samples and No of offsets

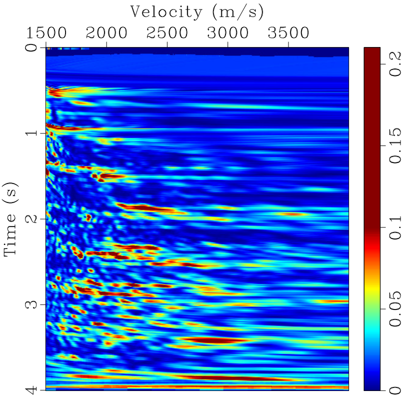
# Velocity analysis



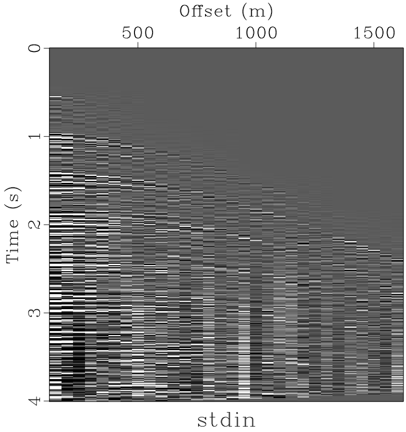
# Velocity analysis



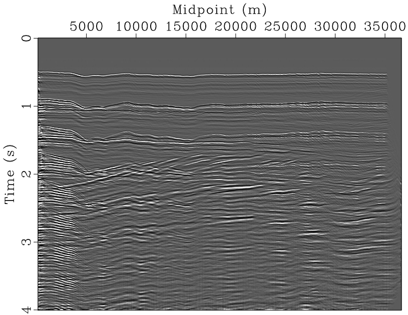
# Velocity analysis



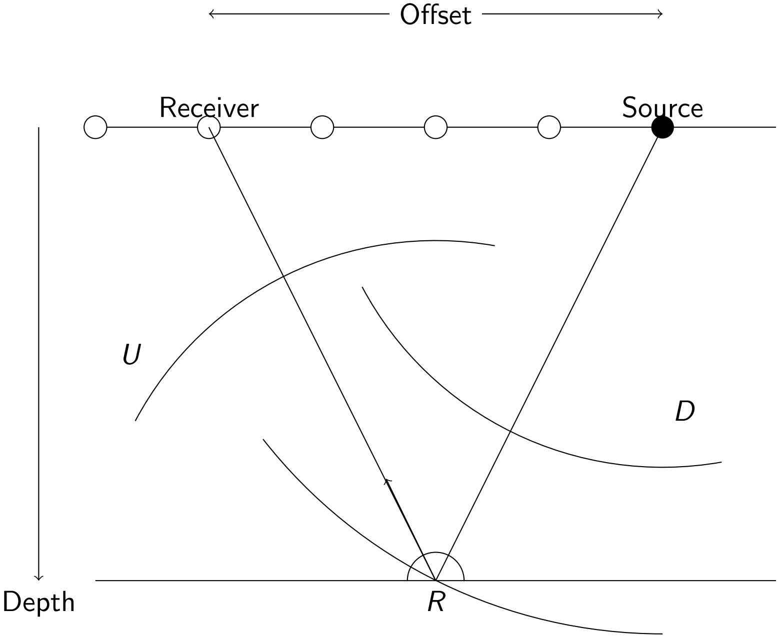
# Velocity analysis



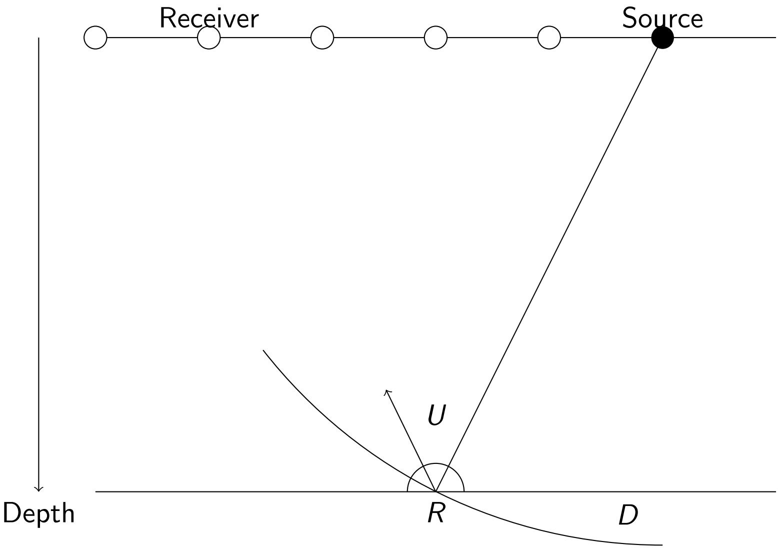
# Velocity analysis



# Migration



# Migration principles



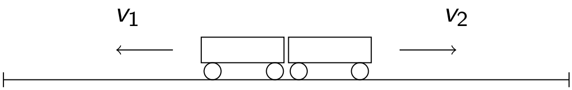
# Migration principles

Make an image by cross-correlating the downgoing and upgoing waves

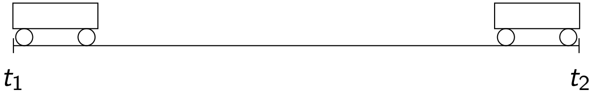
$$R(\mathbf{x}) = \int dt U(\mathbf{x}, t) D(\mathbf{x}, t), \tag{8}$$

where $R$ is the reflectivity and $\mathbf{x} = (x, y, z)$ denotes a position in space, while $t$ is the time.

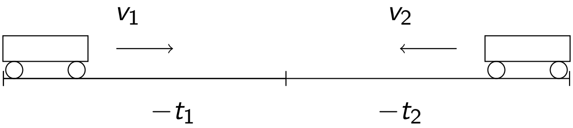
# Time reversal 1



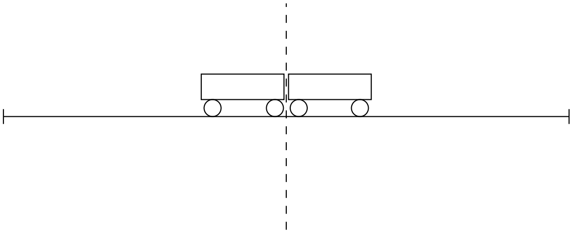
# Time reversal 2



# Time reversal 3



# Time reversal



# Seismic imaging

Forward modeling

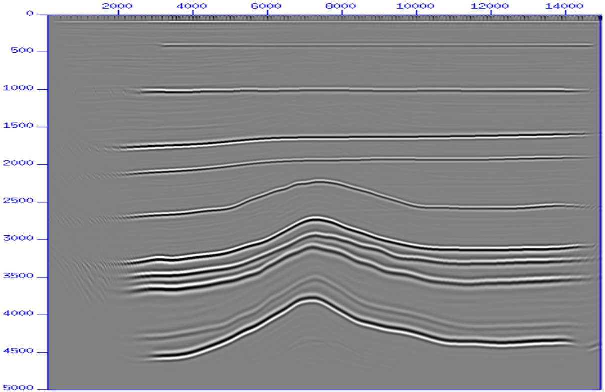
# Time reverse modeling

Backward modeling

# Cross correlation

Imaging

# Complete image



# Kirchhoff migration

Simplest approach for up- and downgoing waves

$$
\begin{aligned}
D(\mathbf{x}, t) &= A\delta(t - \tau_s), \\
U(\mathbf{x}, t) &= BP(t + \tau_r),
\end{aligned} \tag{9}
$$

where

- $\mathbf{x}$: reflection point,
- $A$ and $B$: amplitude factors
- $P$: Data at the surface
- $\tau_s$ traveltime from the source to the reflection point
- $\tau_s$ traveltime from the receiver to the reflection point

# Kirchhoff migration

Using equation (11) and equation (10), I get

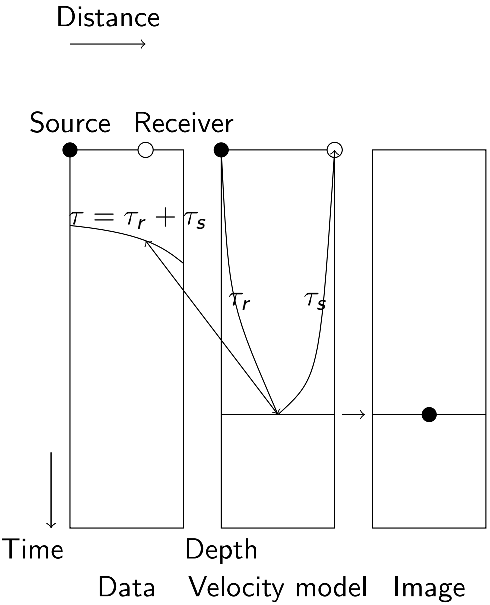
$$R(\mathbf{x}) = \int dt\, A\delta(t-\tau_s)BP(t+\tau_r), \tag{10}$$

which after integration gives

$$R(\mathbf{x}) = ABP(\tau_s+\tau_r). \tag{11}$$

If we disregard the amplitude factor $AB$ the image is simply equal to the amplitude of the recorded data at a time equal to $\tau = \tau_r + \tau_s$.

# Kirchhoff migration



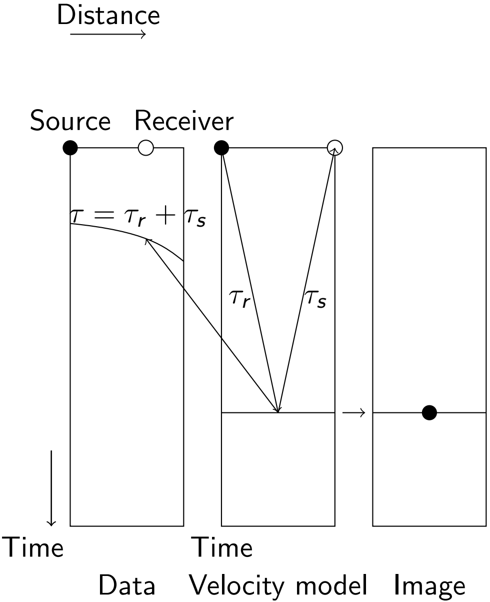
# Kirchhoff migration

Many source-receiver pairs will contribute to the same imaging point, so that equation (13) becomes

$$R(\mathbf{x}) = \sum_{r,s} P(\tau_r + \tau_s), \tag{12}$$

where we have neglected the amplitude factors and the summation is over all source-receiver pairs contributing to the image point.

# Kirchhoff migration



# Kirchhoff migration

Assume velocity $c(\mathbf{x})$ is constant, then

$$\begin{aligned}
\tau_s &= \sqrt{\frac{(x-x_s)^2+(y-y_s)^2}{c^2}+\tau_0^2}, \\
\tau_r &= \sqrt{\frac{(x-x_r)^2+(y-y_r)^2}{c^2}+\tau_0^2},
\end{aligned} \tag{13}$$

- $(x_r, y_r)$, $(x_s, ys)$: Source, receiver positions
- $\tau_0 = z/c$ :vertical traveltime and
- $c$ : velocity.

# Kirchhoff migration

We then get for the total traveltime $\tau$

$$\tau = \sqrt{\frac{(x - x_s)^2 + (y - y_s)^2}{c^2} + \tau_0^2} + \sqrt{\frac{(x - x_r)^2 + (y - y_r)^2}{c^2} + \tau_0^2}. \quad (14)$$

The image $R$ is then

$$R(x, y, \tau_0) = \sum_{r,s} P(\tau_s + \tau_s), \quad (15)$$

# Zero-offset simple migration

We get for zero-offset (stack) data (2D)

$$\tau = 2\sqrt{\frac{(x - x_s)^2}{c^2} + \tau_0^2}. \tag{16}$$

# Zero-offset simple migration

Migration of zero-offset section can then be done as:

- Chose a point $x_s, \tau_0$ on the section.
- Loop over all $x$ values, compute $\tau$ and sum all $P(\tau)$ (Summing over an hyperbola with center at $x_s, \tau_0$.)
- Put the sum at location $x_s, \tau_0$ in the output.
- Repeat for all possible points $x_s, \tau_0$.

# Basic+ processing sequence

1. Input data
2. Velocity analysis
3. Migration
4. Output result